

Course layout and initial setup

R0005E - Measurement and Feedback Control

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Course team



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Setup and changes

Basically:

- Focus of the course is on modeling and control
- Since measurements are essential to feedback control it is also dealt with in the course

Changes since 2023-LP2

- We revert back to classic lectures, based on the feedback we have received
- Video lectures of the theoretical concepts are also available as complementary material and for self-reading.
- The study guide is updated to reflect the changes
- The theory part is now taught in the beginning and 3 of the problem solving sessions are aligned with that.
- Lab assignments are now condensed towards the end where the students can focus on the practical part for 2 weeks
- The lab assignments are restructured and the difficulty is reduced.
- To get started with Matlab and Simulink you have 4 weeks to get prepared before the first lab assignment.

Efforts in the course

- The course is designed for a 50% work load, which equals to 20 hours per week for an average student.
- If the content is difficult to grasp you might need to put down more efforts than that. If you find the material easy, it might be less.
- The average work load is distributed as follows:
 - Theory (3.5 credits): approx. 9 hours/week
 - Problem solving (2 credits): approx. 5.5 hours/week
 - Laboratory work (2 credits): approx. 5.5 hours/week
- Splitting efforts is now easier as the laboratory work is focussed on 2 calendar weeks.
- The theory part still requires you to prioritize between reading the literature and solving theory problems. If you need help with that please contact us.

How to work in the course:

- Read the study guide
- Plan your activities according to the time line in the study guide and schedule.
- Get started right away - which means TODAY!
- Do not try to solve all recommended problems. Do the ones referred to in the videos/slides first. If you have some time left, pick some more.
- Read the book together with the slides and if you like with the lecture videos.

Laboratory work = 2 credits

- Level control setup is common in many industries even in space applications
- You are required to design, implement and **demonstrate** a closed loop control system.
- The work is done in Matlab/Simulink. Block diagram based programming.
- We are your customers and provide you with a performance specification.



- Lab assignments A and B contain many instructions, while C requires more independent work.
- It is expected that you finish the lab assignment during the supervised session, but for that you need to be prepared.
- Each assignment manual contains a preparation section, make sure you have performed these tasks with your group.
- If you do not prepare, you might not be able to finish the assignment and time and you need to put extra time in the lab.
- We are providing supervision in the lab during the scheduled times, but you are always welcome to our offices to ask questions, arrange an online meeting, or send an email with questions.

A tricky process

- The lab process is designed to mimic a real-life process
- The process exhibits variations which are common in real-life, but not usual for a lab
- You will benefit from that in your learning, but you might get confused at times
- Do not hesitate to ask!

Problem solving = 2 credits

General setup

- Four problem solving sessions with seven (7) problems in each session.
- The problem solution is presented by the students during the session

How is it conducted?

- Individual hand-in of solutions to the problem sets prior to the session.
- Participation in the four (4) problem solving sessions is mandatory.
- We randomly select a student to present the solution of a problem.

How to get the credits?

- You need to solve at least six (6) out of the seven (7) problems in each session.
- A total number of 25 problems need to be solved to pass the problem solving.
- You are required to hand in an extra problem set, if you
 - fail to submit solutions in time, or
 - fail to solve sufficient number of problems, or
 - refuse to present a solution when you are select

Schedule

Slot	Week	Date	Start	End	Room, Location	Activity	Information
1	w36	2024-09-02	14:45	16:15	E1024	Lecture	
2	w36	2024-09-04	08:15	09:45	E1024	Lecture	
3	w36	2024-09-05	10:15	11:45	E1024	Lecture	
4	w37	2024-09-09	13:00	14:30	E1026	Lecture	
5	w37	2024-09-11	10:15	11:45	E1024	Lecture	
6	w37	2024-09-13	13:00	14:30	E1024	Lecture	
7	w38	2024-09-16	14:45	16:15	E1026	Lecture	
8	w38	2024-09-17	10:15	11:45	E1024	Problem solving	
9	w38	2024-09-18	08:15	09:45	E1024	Lecture	
10	w38	2024-09-19	10:15	11:45	E1024	Lecture	
11	w39	2024-09-23	08:15	09:45	E1024	Lecture	
12	w39	2024-09-25	10:15	11:45	E1024	Lecture	
13	w39	2024-09-26	14:45	16:15	E1024	Lecture	
14	w39	2024-09-27	10:15	11:45	E1024	Problem solving	
15	w40	2024-09-30	14:45	16:15	E1026	Lecture	
16	w40	2024-10-02	08:15	09:45	E1024	Lecture	
17	w40	2024-10-03	10:15	11:45	E1024	Lecture	
18	w40	2024-10-04	10:15	11:45	E1026	Problem solving	
19	w41	2024-10-07	13:00	14:30	A1545	Lecture	
20	w41	2024-10-08	08:15	09:45	A1521, A1522	Laboration	Lab A - Track 1
	w41	2024-10-08	10:15	11:45	A1521, A1522	Laboration	Lab A - Track 1
	w41	2024-10-08	14:45	16:15	A1521, A1522	Laboration	Lab A - Track 2
	w41	2024-10-08	16:30	18:00	A1521, A1522	Laboration	Lab A - Track 2
21	w41	2024-10-10	13:00	14:30	A1521, A1522	Laboration	Lab B - Track 1
	w41	2024-10-10	14:45	16:15	A1521, A1522	Laboration	Lab B - Track 1
	w41	2024-10-11	13:00	14:30	A1521, A1522	Laboration	Lab B - Track 2
	w41	2024-10-11	14:45	16:15	A1521, A1522	Laboration	Lab B - Track 2
22	w42	2024-10-15	08:15	09:45	A1521, A1522	Laboration	Lab C - Track 2
	w42	2024-10-15	10:15	11:45	A1521, A1522	Laboration	Lab C - Track 2
	w42	2024-10-15	14:45	16:15	A1521, A1522	Laboration	Lab C - Track 1
	w42	2024-10-15	16:30	18:00	A1521, A1522	Laboration	Lab C - Track 1
23	w42	2024-10-18	08:15	09:45	A1521, A1522	Laboration	Lab C - Track 2
	w42	2024-10-18	10:15	11:45	A1521, A1522	Laboration	Lab C - Track 1
24	w43	2024-10-21	13:00	14:30	E1024	Lecture	
25	w43	2024-10-22	08:15	09:45	A1521, A1522	Demonstration	
	w43	2024-10-22	10:15	11:45	A1521, A1522	Demonstration	
26	w43	2024-10-22	14:45	16:15	A1547	Problem solving	